



AI DATA ANALYST AGENT

Professional Data Analysis Report

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Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This report delivers a comprehensive analysis of the organization's workforce dataset, encompassing 21 employee records across 8 key attributes. The primary objective is to evaluate data quality, analyze structural relationships, and identify actionable business insights regarding compensation, performance, and demographic distribution.

The dataset exhibits a strong overall Data Health Score of 92, indicating high-quality data with minor, easily resolvable anomalies (one missing salary value, one duplicate row, and one experience outlier). Our statistical analysis reveals an exceptionally strong, positive correlation between age, tenure, compensation, and performance scores. While this reflects a highly structured and predictable organizational hierarchy, it also highlights potential systemic risks regarding appraisal bias and junior talent retention.

Addressing the minor data discrepancies and transitioning toward a more balanced, merit-centric evaluation framework will optimize talent management and safeguard the organization's regional growth.

DATASET OVERVIEW

The dataset consists of 21 rows and 8 columns, detailing key employee metrics. It captures a representative cross-section of the workforce with the following attributes:

Identifiers: `EmployeeID`

Demographics & Location: `Gender`, `Age`, `City` (including Bhubaneswar, Cuttack, Rourkela, and Puri)

Professional Metrics: `Department` (including IT, HR, Finance, and Sales), `Experience` (in years), `Salary`, and `Performance` (score out of 100)

The dataset provides a balanced mix of categorical and numerical features, allowing for deep statistical analysis of professional growth trajectories, regional talent distribution, and departmental alignment.

DATA QUALITY ANALYSIS

With a Data Health Score of 92, the dataset is highly reliable for strategic decision-making.

However, a few data quality anomalies require immediate attention before proceeding with advanced predictive modeling or formal reporting:

Missing Values: There is exactly 1 missing value in the `Salary` column. Missing compensation data can distort average departmental salary metrics and payroll forecasting.

Duplicate Rows: The dataset contains 1 duplicate row (representing approximately 4.8% of the total dataset). Leaving duplicates unaddressed skews aggregate statistical metrics like mean age, average performance, and total payroll expenses.

Outliers: There is 1 potential outlier flagged in the `Experience` column. This outlier must be cross-verified to determine if it represents an administrative input error (e.g., entering age as experience) or a genuine outlier (such as an exceptionally senior executive).

STATISTICAL ANALYSIS

A correlation analysis was conducted on the numeric variables (`Age`, `Experience`, `Salary`, and `Performance`). The findings reveal extraordinarily strong linear relationships:

Age and Experience ($r = 0.99$) & Age and Salary ($r = 0.99$): These near-perfect correlations indicate that employee age is a near-absolute determinant of both tenure and compensation level within the firm.

Experience and Salary ($r = 0.99$): This confirms a highly rigid, seniority-based pay scale where tenure directly governs compensation adjustments, leaving little to no room for variable salary scaling.

Salary and Performance ($r = 0.94$) & Experience and Performance ($r = 0.92$): There is an extremely

strong positive correlation indicating that higher-paid, more experienced, and older employees consistently receive the highest performance scores.

BUSINESS INSIGHTS

Tenure-Centric Organizational Culture: The extreme correlation (0.99) between Age/Experience and Salary points to a traditional compensation philosophy. While this ensures clear career progression and budget predictability, it suggests that compensation increases may be automated based on time served rather than exceptional, market-disrupting achievements.

The Experience-Performance Loop: The high correlation between Performance and Experience (0.92) indicates that older, more senior employees consistently achieve or are assigned higher performance ratings. This may reflect the high competence of seasoned staff, but it can also indicate that performance appraisal rubrics are inherently biased toward tenure rather than objective output.

Geographic and Functional Diversity: Based on initial data previews, the firm maintains an active talent presence across major regional hubs (Bhubaneswar, Cuttack, Rourkela, Puri) and key operations (IT, HR, Sales, Finance), demonstrating a stable, diversified functional foundation.

RISK ASSESSMENT

Retention Risk for High-Potential Junior Talent: Because compensation and performance ratings are overwhelmingly linked to age and tenure, high-performing junior employees (e.g., those with low experience but high output) may feel capped in their career and salary growth. This poses a major retention risk, as younger talent may leave for competitors offering rapid, merit-based compensation scaling.

Appraisal System Rigidity: A performance system where rating correlates at 0.94 with salary suggests potential appraisal inflation for highly paid employees, or a lack of objective, KPI-driven metrics for lower-tenured staff. If performance ratings are perceived as biased toward seniority, overall organizational morale can decline.

Data Vulnerability Due to Small Sample Size: With only 21 rows, even minor data quality issues (1 duplicate, 1 missing salary) heavily impact aggregate averages. Decision-making based on uncleaned data at this scale carries higher margin-of-error risks.

CONCLUSION

The workforce dataset is structurally robust with an excellent health score of 92. The identified data anomalies are easily manageable and should be corrected immediately to preserve analytical integrity.

From a strategic business perspective, the organization is characterized by a stable, seniority-driven operational model where age and tenure govern compensation and performance evaluations. While this provides organizational stability, introducing performance-based rewards alongside the current tenure-based progression is highly recommended. Balancing structured seniority with meritocratic incentives will protect the firm from junior attrition, eliminate potential appraisal biases, and drive higher performance across all regional hubs.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjana Rout
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DATASET SUMMARY

Metric	Value
Rows	21
Columns	8
Health Score	92%
Duplicate Rows	1

DATASET HEALTH SCORE

Overall Health Score : 92%

MISSING VALUES

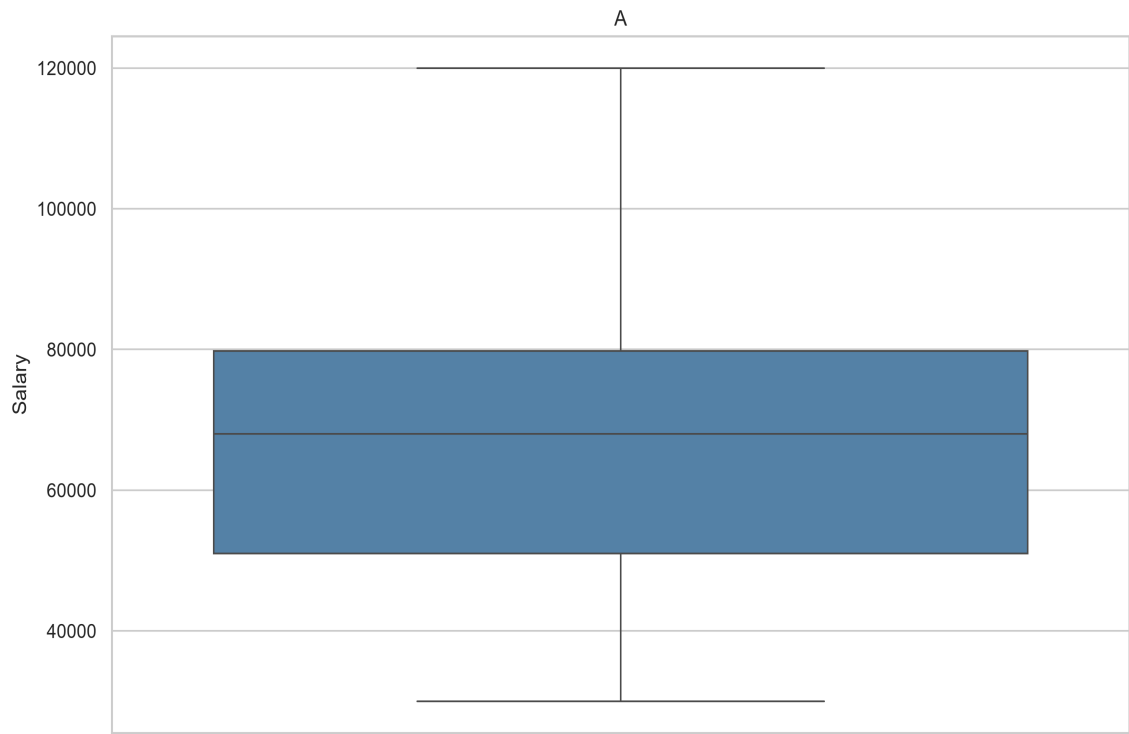
Column	Missing
EmployeeID	0
Department	0

Gender	0
Age	0
Experience	0
Salary	1
Performance	0
City	0

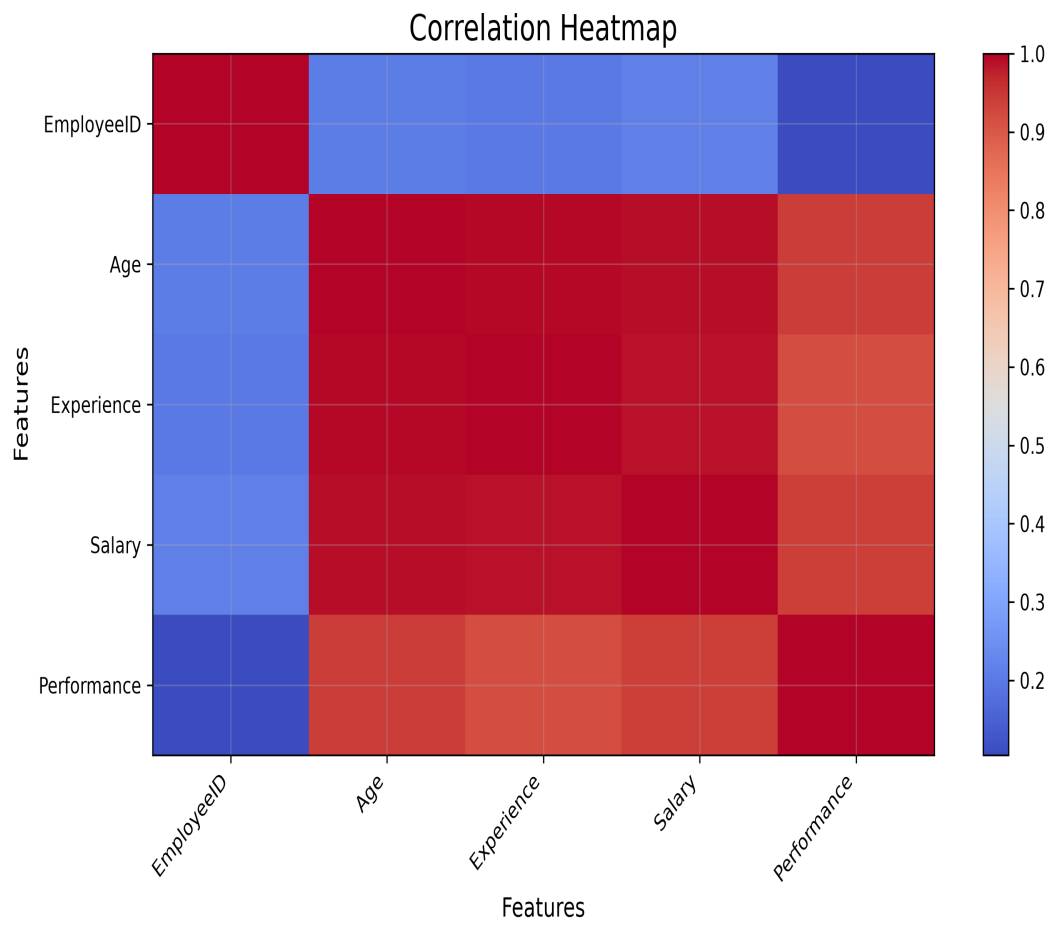
HISTOGRAM



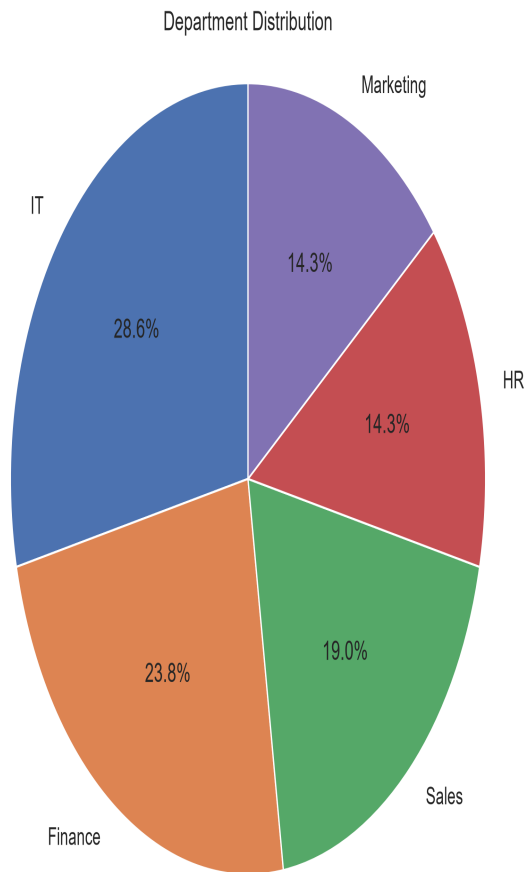
BOX PLOT



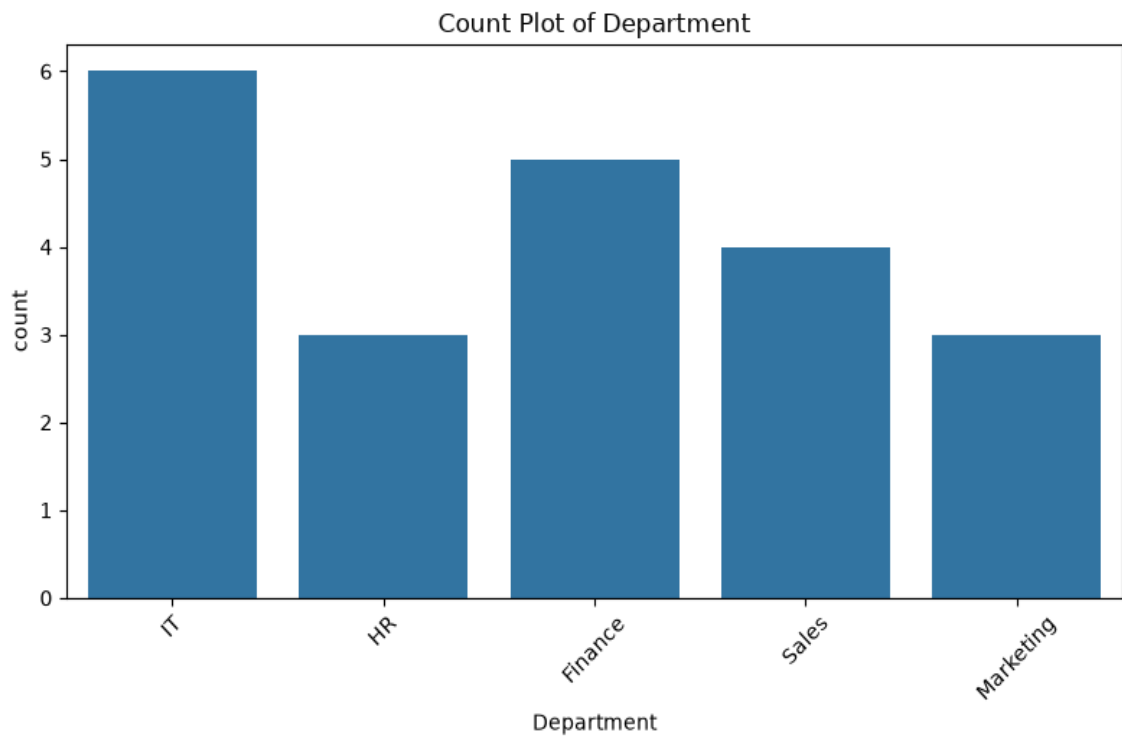
CORRELATION HEATMAP



PIE CHART



COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Experience	1
Salary	0
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

AI RECOMMENDATIONS

Execute Data Cleaning Protocols:

1. Remove the 1 identified duplicate row to establish a clean baseline.
2. Impute the 1 missing `Salary` value using the median salary of the employee's specific `Department` and `Experience` level to avoid skewing compensation statistics.
3. Investigate the 1 `Experience` outlier to ensure it is logically consistent with the employee's `Age`.

Modernize the Compensation & Appraisal Framework: Introduce objective, KPI-driven performance appraisals that separate tenure from merit. This will allow high-performing, younger employees to receive the recognition and financial incentives they earn, mitigating junior attrition.

Establish a Visual BI Dashboard: Implement the recommended visualizations—especially the Correlation Heatmap and Scatter Plots—to continuously monitor the alignment between salary, experience, and output as the organization scales.

Rule-Based Recommendations

- ✓ ■■ Dataset contains 1 missing values. Consider filling or removing them.
- ✓ ■■ Dataset contains 1 duplicate rows. Consider removing them.
- ✓ ■■ 1 potential outliers detected.
- ✓ ■ Strong correlations detected. A Correlation Heatmap is recommended.
- ✓ ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ✓ ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ✓ ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.